

Synchronous FIFO

IP Design Documentation

Version v1.0

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1. Revision

Version	Date	Author	Description
v1.0	August 2026	Dat Tran Tan	Initial release

2. Overview

2.1 Purpose

The synchronous FIFO provides temporary data storage between a producer and a consumer operating in the same clock domain. It supports parameterizable data width and FIFO depth.

2.2 Features

- Single-clock synchronous FIFO
- Parameterizable data width and depth
- Full and empty status flags with FIFO usage monitoring
- Asynchronous reset and synchronous flush

3. Architecture

3.1 Block Diagram

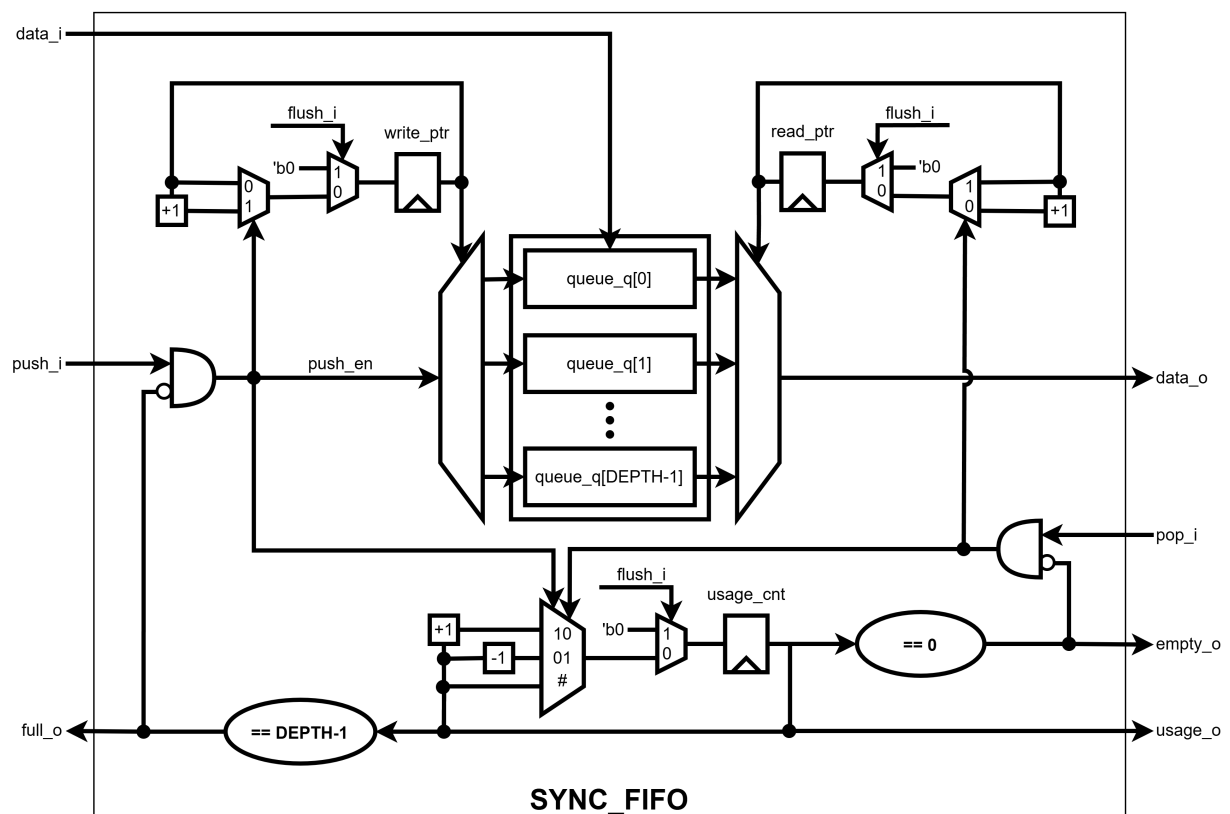


Figure 1: Synchronous FIFO block diagram

3.2 IO Ports

Group	Port	Direction	Width	Description
Clock/Reset	clk_i	Input	1	Clock signal
	rst_i	Input	1	Asynchronous reset
Write Source	push_i	Input	1	Write request
	data_i	Input	DATA_WIDTH	Write data
Read Destination	pop_i	Input	1	Read request
	data_o	Output	DATA_WIDTH	Read data
Status	full_o	Output	1	FIFO full status
	empty_o	Output	1	FIFO empty status
	usage_o	Output	USAGE_WIDTH	FIFO occupancy

Table 1: Synchronous FIFO interface

3.3 Config Parameters

Parameter	Default	Range	Description
DATA_WIDTH	32	> 0	Data width
DEPTH	16	> 1	FIFO depth

Table 2: Configurable parameters

4. Functional Description

The synchronous FIFO provides temporary storage between a write source and a read destination operating in the same clock domain. Data is written when **push_i** is asserted and the FIFO is not full, and is read when **pop_i** is asserted and the FIFO is not empty.

4.1 Write Operation

A write operation is performed on the rising edge of **clk_i** when **push_i** is asserted and **full_o** is deasserted. The input data on **data_i** is stored at the current write location.

4.2 Read Operation

A read operation is performed on the rising edge of **clk_i** when **pop_i** is asserted and **empty_o** is deasserted. The next stored data is provided on **data_o**.

4.3 Reset Behavior

Asynchronous assertion of **rst_i** resets the FIFO state, including the read and write pointers and status indicators. After reset, the FIFO is empty.

4.4 Flush

A synchronous flush clears the FIFO contents and restores the FIFO to the empty state on the rising edge of **clk_i**.

4.5 Status

The **full_o** and **empty_o** outputs indicate whether the FIFO can accept a write or provide a read, respectively. The **usage_o** output indicates the current FIFO occupancy.

5. Verification Coverage

5.1 Verification Environment

5.2 Test Cases

Test	Status	Description
Reset test	PASS	Verify reset behavior
Write test	PASS	Verify FIFO write operation
Read test	PASS	Verify FIFO read operation
Full test	PASS	Verify full condition
Empty test	PASS	Verify empty condition
Boundary test	PASS	Verify boundary transitions

5.3 Coverage Summary

Metric	Result
Functional Coverage	–
Code Coverage	–
Assertion Coverage	–
Test Cases	6/6 PASS

6. Synthesis Report & SDC

6.1 Synthesis Configuration

Item	Value
Technology	–
Library	–
Tool	–
Clock	–
Target Frequency	–

6.2 Synthesis Results

Metric	Result
Cell Area	–
Sequential Area	–
Combinational Area	–
Worst Slack	–
Total Power	–

6.3 SDC

```
create_clock -name clk \
    -period <PERIOD> [get_ports clk]

set_input_delay <VALUE> \
    -clock clk [all_inputs]

set_output_delay <VALUE> \
    -clock clk [all_outputs]
```

7. Note